

## Technical Note

### SURFACE PRIMERS

#### DESCRIPTION

Dorf Ketal Tyzor® organic titanates make excellent primers for the treatment of a variety of metals, plastics, and even fluoropolymers to accept paints, inks, rubber-based coatings and many others too numerous to mention.

#### METALS

To use Tyzor® organic titanates as a primer, dilute to a concentration of 0.1 to 2% and apply as a thin coating to the surface of the substrate. As the solvent evaporates, atmospheric moisture hydrolyzes the titanate to form a thin, reactive film of polymeric  $\text{TiO}_2$ . This film bonds tenaciously to the substrate and to the subsequent coating through functional groups containing active hydrogen. Usually the more reactive alkoxy titanates, such as Tyzor® TPT and TNBT, are used; however, chelated titanates can be used if heat is available to accelerate the hydrolysis. If the solution is applied too heavily or dried in an atmosphere too high in humidity, the titanate may powder on the surface rather than form a continuous film. A relative humidity of 30-60% is recommended.

Titanates are used as primers for coatings such as paints, inks, and silicone rubbers on steel, aluminum, zinc, and copper to promote adhesion and to improve resistance to corrosion, water, and discoloration.<sup>1-11</sup>

#### PLASTICS

Tyzor® organic titanates also are used to promote adhesion of printing inks, adhesives, paints, and polymeric resins to a wide range of plastic films and sheets.<sup>12-22</sup> Titanates are particularly effective when the plastic has been "surface-treated" to provide bonding sites.

Suitable plastics include Mylar® [poly(ethylene terephthalate)], polyolefins, polyimides, nylon, polyurethanes, epoxies, phenolics and silicones. Applications include improved adhesion of inks to plastic films and improved adhesion of laminates used in packaging material and automotive parts.

#### SILICONE ADHESIVES

Organic titanates are used as primers to promote adhesion of silicone rubber adhesives to metal, plastic, glass, ceramics, concrete, wood and fabrics.<sup>23-29</sup>

Silicone adhesives are used in the building, aviation, and automotive industries.

#### FLUOROPOLYMERS

Titanates are used as primer coatings to improve adhesion of resins, varnishes, inks and dyes to poly-tetrafluoroethylene (PTFE) surfaces.<sup>30</sup> After treatment with Tyzor® TPT, PTFE surfaces demonstrate decreased slip and improved wetting by inks, resins and adhesives.<sup>31</sup> Tyzor® GBA is used as a primer coating to promote the adhesion of glass cloth to PTFE.<sup>32</sup>

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